densol

On-chain compression for Solana Anchor programs

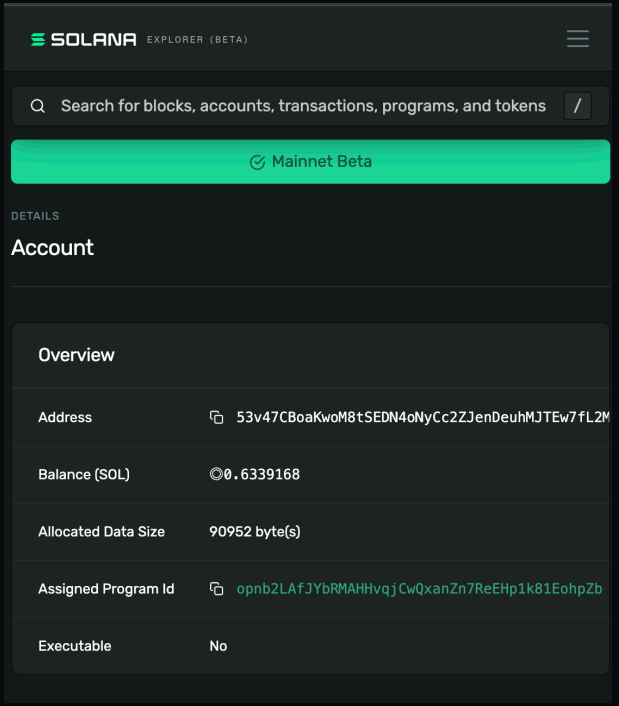
Up to 54× compression · Data stays on-chain · Off-chain infrastructure not required

Blockchain Hack Warsaw · March 22, 2026

// PROBLEM

Solana rent is proportional to account size

- Every byte stored on-chain locks SOL as rent
- Data-heavy programs pay enormous, compounding costs
- OpenBook pre-allocates 90,952 bytes per orderbook side



** Measured on mainnet data account*

// PROBLEM

Solana rent is proportional to account size (json-like data)

1 KB	10 KB	90 KB	1 MB	4 MB	10 MB (MAX)
0.008 SOL	0.072 SOL	0.634 SOL	7.30 SOL	29.19 SOL	72.98 SOL
~$0.70	~$6.30	~$55	~$637	~$2,549	~$6,372
Compression					
?	?	?	?	?	?

// SOLUTION

densol - LZ4 that fits the SBF VM

no_std Rust

Zero libc dependencies compiles for BPF

16 KB stack table

Hash table lives on the stack; heap stays free for your data

ChunkedLz4<4096>

4 KB chunks decompress one at a time, no heap ceiling

#[compress]

Single derive attribute on any Vec<u8> field in an Anchor account

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// INSIGHT

Data stays on-chain.
No Merkle trees. No indexer.

ZK Compression

- ~292,000 CU per tx (fixed overhead)
- Data lives off-chain in ledger/indexer
- Requires Helius Photon indexer
- Great for NFTs & millions of tiny accounts

densol

- ~10,000–15,000 CU per write (for accounts under 10 KB)
- Data stays fully on-chain
- Zero off-chain infrastructure
- Drop-in for any Anchor/Borsh program

// RESULTS

How much is your account costing you? (json-like data)

1 KB	10 KB	90 KB	1 MB	4 MB	10 MB (MAX)
0.008 SOL	0.072 SOL	0.634 SOL	7.30 SOL	29.19 SOL	72.98 SOL
~$0.70	~$6.30	~$55	~$637	~$2,549	~$6,372
Compression (structured / sparse data)					
150 B (6.83×)	186 B (55.05×)	3,874 B (23.79×)	43,212 B (24.27×)	172,829 B(24.27×)	432,073 B(24.27×)
0.001935 SOL	0.002185 SOL	0.027854 SOL	0.301646 SOL	1.203781 SOL	3.008 SOL

// MARKET

Show every protocol the SOL they're leaving on the table

01

Public indexer dashboard

Scans top protocols. Shows each team: X inactive accounts, Y SOL recoverable. They see their number before we call.

02

densol library (free, open source)

Protocol integrates themselves. cargo add densol. Single #[compress] attribute. Works with Anchor account.

03

Managed SaaS

Auto-compress inactive accounts on behalf of integrated protocols.

densol is alive.

Use it today.

@JAKZYG

cargo add densol

crates.io/crates/densol

github.com/ZygmuntJakub/densol

benchmarks · source · docs at densol.dev